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Language-Guided Active View Selection for Vision-Language Perception

by

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Abstract

Robotic perception often requires identifying not only what object is present, but which viewpoint best satisfies a task-specific natural language instruction. Different views of the same object reveal different visual features such as handles, printed designs, or affordance regions. Under a limited observation budget, a system must therefore select a small number of informative views rather than inspect every candidate.

This thesis investigates language-guided active view selection for multi-view object perception. The system starts from a random initial view and sequentially selects additional views under budgets $k \in \{1, 2, 3, 4, 5\}$. After the budget is exhausted, an observed-view reranker returns a final prediction. The work evaluates a CLIP relevance-plus-novelty active policy, a learned MLP reranker trained on frozen CLIP features, and a trained active view selector that uses the same CLIP+MLP architecture. These methods are compared with random selection and exhaustive CLIP ranking.

Two datasets are used. The first is a CO3D-based multi-view annotation set covering five object categories with 30 instances each and five viewpoints per instance. Ground-truth views are instruction-dependent and view order is shuffled to remove first-view bias. The second is derived from the UMD+GT RGB-D affordance dataset. Affordance masks are converted into language-guided view-selection samples, where ground-truth views are those whose affordance area is at least 80% of the largest area among the five candidates.

On the shuffled CO3D evaluation set, learned reranking is the main source of improvement: at budget 3, random selection with learned reranking reaches 0.484 accuracy, while the full trained-selection plus learned-reranking pipeline reaches 0.520, compared with 0.310 for the random-plus-CLIP baseline. On UMD+GT, evaluated over 20 random-initial trials, the full pipeline is clearly strongest, reaching 0.669 ± 0.011 already at budget 2 and 0.680 ± 0.008 at budget 3, against about 0.46 for random-plus-CLIP. Ablation studies show that the learned reranker contributes most of the gain, while the trained selector is especially useful when the dataset contains smoothly varying viewpoint cues, as in UMD affordance sequences.

Overall, this thesis shows that language-guided scoring can support active view selection, that a lightweight MLP on frozen CLIP features substantially improves final view ranking, and that a trained selector can further improve low-budget performance

when viewpoint variation is informative. Remaining challenges include overlapping selector/reranker training objectives, optimistic access to unseen-view features during selection, and limited sample counts for some affordance categories.

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Abbreviations

BCE Binary Cross-Entropy

CLIP Contrastive Language–Image Pre-training

CO3D Common Objects in 3D

GT Ground Truth

MLP Multi-Layer Perceptron

RGB-D Red, Green, Blue, Depth

UMD University of Maryland

VLM Vision-Language Model

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Chapter 1

Introduction

1.1 Motivation

Robotic perception often requires more than recognising an object category from a single image. In many practical tasks, a robot must interpret an object according to a specific instruction. For example, it may need to find the view where a cup handle is clearly visible, where a printed design is front-facing, or where a grasping region of a tool is most exposed. These tasks require the system to connect visual information with natural language.

A major challenge is that objects are three-dimensional, and different viewpoints reveal different visual details. A side view may expose a handle, while a frontal view may expose a printed pattern. In affordance-centred settings, different viewpoints may also change how clearly a graspable, cutting, scooping, or pounding region appears. Therefore, choosing the right view is an important part of perception.

In real robotic systems, inspecting every possible view is often inefficient. A robot may have limited time, computation, camera movement range, or energy. This motivates active perception, where the system selects informative observations instead of passively using all available inputs. This thesis focuses on language-guided active view selection:

given a natural language instruction and a set of candidate views, the system selects a limited number of views under an observation budget and then ranks the observed views to produce a final answer.

Earlier experiments in this project showed that a CLIP-based relevance-plus-novelty policy can outperform non-semantic baselines such as random selection, but that much of the useful signal comes from image–text similarity rather than from the selection policy itself. This thesis therefore extends the pipeline in three ways. First, it builds larger and better-balanced evaluation datasets, including a shuffled CO3D subset and an affordance-derived UMD+GT subset. Second, it trains a learned MLP reranker on frozen CLIP features. Third, it trains an active view selector with the same architecture and compares both components through ablation studies.

1.2 Problem Statement and Research Questions

The problem addressed in this report is language-guided active view selection for multi-view object perception. Given a set of candidate views of an object and a natural language instruction, the system must select a limited number of views and return a final predicted view that is most likely to satisfy the instruction.

For an object with candidate views

$$V = \{v_1, v_2, \ldots, v_n\}$$

and a language instruction q, the system first selects an observed subset

$$S_k \subseteq V$$

where k is the observation budget. It then ranks the views in S_k and returns the top-ranked view $\hat{v}$. The prediction is correct if $\hat{v}$ belongs to the ground-truth set $G \subseteq V$.

The main research question is:

> Can a language-guided active view selection strategy, together with a learned observed-view reranker, improve vision-language perception under a limited observation budget?

This main question is divided into the following sub-questions:

- How can natural language instructions be used to score and rank candidate object views?
- Does a CLIP-based active policy outperform random selection and exhaustive CLIP ranking?
- Can a learned MLP reranker improve final view prediction beyond CLIP cosine similarity?
- Can a trained active view selector improve over a fixed CLIP relevance-plus-novelty policy?
- How do these methods transfer from CO3D-style object-view instructions to affordance-derived UMD+GT samples?
- Which pipeline components contribute most to accuracy gains under different budgets?

The central hypothesis is that language-guided image–text similarity provides a useful prior for view selection, but that a lightweight learned model can better map CLIP features to instruction-specific ground-truth views. A secondary hypothesis is that a trained selector can further improve low-budget performance when viewpoint variation is informative, as in turntable affordance sequences.

1.3 Research Methodology

The project follows an experimental computer science methodology and is organised into six stages.

First, the task is formulated as limited-budget sequential view selection followed by observed-view reranking. Each sample contains five candidate views, a language instruction, and one or more ground-truth views.

Second, related literature is reviewed in active perception, view planning, vision-language models, CLIP-based image–text matching, multi-view object datasets, and language-guided active perception.

Third, two evaluation datasets are prepared. The CO3D-based set contains five categories with 30 object instances each. Views are sampled at 20% increments and later shuffled to remove first-view bias. The UMD+GT-based set converts affordance masks into language-guided samples, defining ground-truth views by relative affordance area.

Fourth, the experimental pipeline is implemented. It includes CLIP feature extraction, a CLIP relevance-plus-novelty active policy, a learned MLP reranker, a trained active view selector, random and exhaustive CLIP baselines, and budget sweeps from 1 to 5.

Fifth, methods are evaluated with repeated random trials. Each trial starts from a random initial view. Accuracy is measured by whether the final top-1 predicted view is ground truth. Top-k ranking accuracy and per-category or per-affordance breakdowns are also reported.

Sixth, ablation studies isolate the contribution of each component relative to a shared baseline of random selection plus CLIP reranking. Gains are reported by category on CO3D and by affordance on UMD+GT.

1.4 Contribution of the Work

This project makes the following contributions:

- It formulates language-guided active view selection as a limited-budget sequential selection and reranking problem.

- It constructs two evaluation settings: a shuffled CO3D multi-view instruction dataset and an affordance-derived UMD+GT view-selection dataset.
- It implements a CLIP-based active policy that combines instruction relevance with viewpoint novelty.
- It trains a learned MLP reranker and a trained active view selector on frozen CLIP text and image features.
- It evaluates all methods under budgets $k \in \{1, 2, 3, 4, 5\}$ with repeated random-initial trials.
- It provides paired ablation studies that measure each component's accuracy gain relative to a random-plus-CLIP baseline.
- It analyses when the full trained pipeline helps, when the learned reranker alone is sufficient, and where remaining limitations remain.

1.5 Structure of the Report

The remainder of this report is organised as follows.

Chapter 2 presents the background and literature review. It introduces active perception, vision-language models, multi-view object perception, language-guided active view selection, and the research gap addressed by this project.

Chapter 3 presents the methodology and implementation. It describes the datasets, CLIP scoring, active policies, learned reranker and selector, baselines, evaluation metrics, and training details.

Chapter 4 presents the experiments and results on CO3D and UMD+GT, including budget curves, top-k analysis, and ablation studies.

Chapter 5 provides the discussion, conclusion, and future work. It summarises the main findings, discusses limitations, and outlines possible directions for extending the

project.

Chapter 2

Background and Literature Review

2.1 Introduction

This chapter presents the background and literature review for the project. The project investigates language-guided active view selection for vision-language perception. The key idea is that a perception system should not always process every available view. Instead, it should select the most informative views according to the current language instruction and observation budget, then rank the observed views to produce a final prediction.

The chapter is organised around five main areas. First, active perception and view planning are introduced as the general motivation for selecting useful viewpoints. Second, vision-language models are discussed as the foundation for connecting images with natural language instructions. Third, multi-view object perception is introduced because the project uses multiple views of the same object. Fourth, language-guided active view selection is reviewed as the closest area to this project. Finally, the chapter summarises the research gap addressed by this thesis.

2.2 Active Perception and View Planning

Active perception refers to the idea that a perception system should actively choose what to observe. Instead of passively receiving visual input, an active system can select camera viewpoints, sensor positions, or image observations to improve task performance. This is especially important in robotics, where perception is often limited by time, computation, occlusion, camera motion, and physical constraints.

Zeng et al. [ZWZL20] provide a survey of view planning in robot active vision. They explain that view planning is concerned with selecting useful sensor viewpoints under task and system constraints. Their survey covers applications such as object reconstruction, scene reconstruction, object recognition, and pose estimation. This is relevant to the present project because language-guided active view selection can be understood as a simplified form of view planning. Instead of moving a physical camera, the system selects from a set of existing candidate views.

The motivation from active vision is that not all views are equally useful. A view may be useful for one task but unhelpful for another. For example, a side view of a cup may be useful for detecting its handle, while a front view may be better for observing a printed design. In tool-affordance settings, a viewpoint may also change how clearly a graspable handle or cutting edge appears. Therefore, active perception requires a task-dependent decision about which viewpoint should be selected.

In this project, the active perception problem is formulated as sequential view selection under a fixed observation budget. The system is given a set of candidate views and must select only a small number of them. After the budget is exhausted, the observed views are reranked and the highest-scoring view is returned as the final prediction. This reflects the practical situation where a robot may not have enough time or resources to inspect every possible view.

2.3 Vision-Language Models and CLIP

Vision-language models connect visual information with natural language. They are useful for tasks where the system must interpret an image according to a text description, question, or instruction. This makes them highly relevant to language-guided perception, where the target visual information is specified by natural language.

A central model for this project is CLIP, introduced by Radford et al. [RKH+21]. CLIP learns visual representations from natural language supervision by training on large-scale image–text pairs. Instead of training only on fixed object categories, CLIP learns to associate images with text descriptions. This enables zero-shot image classification and image–text matching.

The key idea used in this project is CLIP-style image–text similarity. Both images and text are encoded into a shared embedding space. If an image matches a text prompt, their feature vectors should be close in that space. This allows the system to compare each candidate view with a natural language instruction and rank the views according to similarity.

For example, if the instruction is

> find the view where the printed design is front-facing

then each candidate view can be compared with this instruction. The view with the highest image–text similarity score is treated as the most relevant view. This makes CLIP suitable for implementing a simple language-guided scoring policy.

However, CLIP also has limitations. It is strong at broad image–text association, but it may be less reliable for fine-grained viewpoint reasoning. In this project, the model must distinguish between different views of the same object. This can be more difficult than recognising the object category itself. For example, several views may all contain a cup, but only one or two views may clearly show the handle or printed design. This limitation motivates the learned MLP reranker studied in this thesis: CLIP features

are retained as a strong frozen representation, while a lightweight classifier is trained to map those features to instruction-specific ground-truth views.

2.4 Multi-View Object Perception

Multi-view object perception studies how multiple views of the same object can support visual understanding. A single object can appear very different from different camera angles. Some viewpoints may show important visual details, while others may hide them.

The CO3D dataset introduced by Reizenstein et al. [RSH+21] is relevant to this project because it provides real-world multi-view object data. The dataset contains object-centric videos from multiple categories and includes many views of real objects. This type of data is useful for studying how object appearance changes across viewpoints. In this thesis, a curated CO3D subset is used with five viewpoints per object and instruction-dependent ground-truth views.

Affordance-centred RGB-D datasets provide a complementary setting. In particular, the UMD+GT dataset [MTFA15] annotates common tools with per-pixel affordance labels such as grasp, cut, scoop, contain, pound, and wrap-grasp, together with keypoints and turntable sequences from many viewpoints. Although the original annotations are pixel-level rather than view-selection labels, they can be converted into language-guided view-selection samples by asking which viewpoint most clearly presents a given affordance. This provides a second evaluation setting where ground truth can be derived systematically from mask area rather than only from manual front-facing judgements.

Multi-view perception is important for this project because the task is not simply to classify an object. Instead, the system must identify which view best satisfies a language instruction. An object may be recognisable from many views, but the instruction may require a particular visual property to be visible. This makes the problem more specific than standard object recognition.

2.5 Language-Guided Active View Selection

The closest research area to this project is language-guided active view selection. In this setting, the system uses a language query, instruction, or question to decide which view should be selected.

Multi-view visual question answering with active viewpoint selection is directly related to this idea [QSS+20]. In that task, a system must answer a visual question by selecting useful viewpoints. This is similar to the present project because both settings require the system to use language to guide visual observation. The difference is that this project focuses on selecting the view that best satisfies an instruction, rather than directly answering a question.

Sripada et al. [SCGG25] propose scene exploration using vision-language models. Their AP-VLM framework combines active perception with a vision-language model to guide robotic exploration and answer semantic queries. This is highly relevant because it connects active perception with language-based reasoning. Their work supports the idea that a vision-language model can help a robot choose better viewpoints in partially observable environments.

Wang et al. [WCL+25] introduce ActiView, a benchmark for evaluating active perception ability in multimodal large language models. This work is useful for the evaluation perspective of the present project. It shows that active perception is becoming an important capability for multimodal models, especially when the system must reason under limited visual information.

These works show that active perception is moving beyond traditional geometry-based view planning. Modern systems increasingly use semantic goals, language instructions, and multimodal reasoning to decide what to observe. This supports the direction of the present project, where a natural language instruction is used to rank and select candidate object views.

2.6 Research Gap

The reviewed literature provides the foundation for this project, but there is still a useful gap for a focused experimental study.

Traditional active vision focuses heavily on geometric tasks such as reconstruction, recognition, and pose estimation [ZWZL20]. Vision-language models such as CLIP provide a way to connect images and text [RKH+21], but they are not specifically designed for fine-grained active view selection. Multi-view datasets such as CO3D provide suitable data for studying object viewpoints [RSH+21], but selecting the best view according to a language instruction remains challenging. Affordance datasets provide rich pixel-level labels, yet they are rarely converted into budgeted language-guided view-selection benchmarks.

Recent work on active perception with vision-language models shows strong potential [SCGG25, WCL+25], but several questions remain open for a practical pipeline:

- How much of the performance comes from CLIP similarity itself, and how much from an active selection policy?
- Can a lightweight learned model on frozen CLIP features improve final observed-view ranking?
- Can a trained selector improve over a fixed relevance-plus-novelty heuristic?
- Do these conclusions hold both on instruction-labelled object views and on affordance-derived view-selection samples?

Therefore, this thesis investigates a focused pipeline: CLIP scoring, heuristic and trained active selection, learned reranking, and paired ablations on CO3D and UMD+GT. The setting is smaller than full robotic exploration, but it isolates an important part of the problem: deciding which views to observe and which observed view best answers a language instruction under a limited budget.

2.7 Summary

This chapter reviewed the main literature related to language-guided active view selection. Active perception and view planning motivate the need to select informative viewpoints under constraints. Vision-language models, especially CLIP, provide a practical method for comparing image views with text instructions. Multi-view object datasets and affordance annotations provide complementary settings where different views contain different useful information. Recent work on active viewpoint selection and VLM-based scene exploration shows that language-guided active perception is a relevant and growing research direction.

Based on this literature, the project formulates language-guided active view selection as a limited-budget sequential selection and reranking problem. The next chapter presents the methodology and implementation used to evaluate this idea.

Chapter 3

Methodology and Implementation

3.1 Introduction

This chapter presents the methodology and implementation of the language-guided active view selection system. The aim of the system is to select the most relevant object views according to a natural language instruction while using a limited observation budget, then return a final predicted view from the observed set.

The project treats active view selection as a two-stage problem. First, a selection policy chooses which views to observe under budget k. Second, a reranker scores the observed views and returns the highest-scoring view as the final answer. The methodology covers dataset preparation for CO3D and UMD+GT, CLIP feature extraction, heuristic and trained selection policies, learned reranking, baselines, evaluation metrics, and training details.

3.2 System Overview

The overall system takes two inputs: a set of candidate object views and a natural language instruction. The candidate views are images of the same object captured from different viewpoints. The instruction describes the visual property or viewpoint that the system should find.

For each test case, the system performs the following steps:

1. Load all candidate views for one sample and the corresponding language instruction.
2. Encode the instruction and each image with a frozen CLIP model.
3. Start from a random initial observed view.
4. While the number of observed views is below the budget, use a selection policy to choose the next view.
5. After the budget is exhausted, score the observed views with a CLIP or learned reranker.
6. Return the top-ranked observed view as the final prediction.
7. Compare the prediction with the ground-truth correct views.

This pipeline separates selection from final ranking. That separation is important because a method may collect useful views yet still fail if the final reranker is weak, or collect redundant views yet still succeed if the correct view happens to be among the observed set and is ranked highly.

3.3 Task Formulation

For each sample, the candidate view set is defined as

$$V = \{v_1, v_2, \ldots, v_n\},$$

where v_i represents the i-th image view and $n = 5$ in all experiments. The natural language instruction is represented as q. Under observation budget k, the selection policy produces an ordered observed set

$$S_k = \{v_{s_1}, v_{s_2}, \ldots, v_{s_k}\}.$$

A reranker then returns

$$\hat{v} = \arg\max_{v \in S_k} \text{Score}(q, v).$$

The prediction is correct if $\hat{v} \in G$, where $G \subseteq V$ is the ground-truth set of acceptable views. Top-k ranking accuracy is also reported by checking whether any of the top-k ranked observed views belongs to G.

3.4 Datasets

3.4.1 CO3D Multi-View Instruction Dataset

The first dataset is derived from CO3D [RSH+21]. Five categories are used: book, cup, handbag, remote, and teddybear. Each category contains 30 object instances, and each instance has five viewpoints sampled at approximately 20% increments of the source sequence. Each sample has a category-specific instruction, for example:

- cup: *find the view where the printed design is front-facing*
- book: *find the view where the book is most front-facing*
- teddybear: *find the view where the teddy bear face is most visible*

Ground-truth views are instruction-dependent and may contain more than one acceptable viewpoint. Early versions of the annotation set were skewed toward first-view correctness. To remove that bias, view order is shuffled within each sample and sample order is shuffled across the dataset. The resulting annotation file therefore places ground-truth views more evenly across the five positions.

The CO3D split follows object-instance numbers:

- evaluation: objects 1–10 per category;
- validation: objects 11–15 per category;
- training: objects 16–30 per category.

This yields 50 evaluation samples, 25 validation samples, and 75 training samples.

3.4.2 UMD+GT Affordance View-Selection Dataset

The second dataset is derived from the UMD+GT RGB-D affordance keypoint dataset [MTFA15]. The source data contains 104 tools across categories such as mug, knife, spoon, hammer, and bowl, with turntable sequences of roughly 200–400 frames per tool. Each frame has an affordance mask with labels for grasp, cut, scoop, contain, pound, and wrap-grasp.

Because the original annotations are pixel-level rather than view-selection labels, a conversion procedure is used:

1. Sample five views at 20% increments of each tool sequence. Multiple shifted start frames produce several five-view sets per tool.
2. For each affordance present on the tool, create one sample with an instruction such as "find the view of the mug where the handle is most clearly visible for grasping".
3. Measure the fraction of image area covered by that affordance in each of the five views.
4. Mark as ground truth every view whose affordance area is at least 80% of the largest area among the five views.

5. Drop samples whose best affordance area is below 0.2% of the image, or where more than three views pass the 80% threshold (no clear best view).

This produces 1,026 samples: 603 train, 78 validation, and 345 test. Splits follow the official UMD instance split for train/test, with 20% of train tools held out for validation. Random-pick chance accuracy is approximately 0.43 because about two of the five views are ground truth on average. Results are grouped by affordance rather than by object category because several categories have few test instances.

3.5 Image and Text Encoding

A frozen OpenCLIP ViT-B/32 model pretrained on OpenAI CLIP weights encodes both images and text. Each candidate image v_i is mapped to an image feature

$$z_i = f_I(v_i),$$

and the instruction q is mapped to a text feature

$$z_q = f_T(q).$$

Features are L2-normalised, so cosine similarity reduces to an inner product:

$$\mathrm{sim}(z_i, z_q) = z_i \cdot z_q.$$

All CLIP parameters remain frozen. Feature caching is used so that each image and instruction is encoded once.

3.6 CLIP Reranking and Heuristic Active Policy

3.6.1 CLIP Reranking

Given an observed set S_k, the CLIP reranker returns the view with the highest similarity to the instruction:

$$\hat{v}_{\mathrm{CLIP}} = \arg\max_{v \in S_k} \mathrm{sim}(f_I(v), f_T(q)).$$

This is the default final-ranking method for the heuristic active policy and for several ablation configurations.

3.6.2 CLIP Relevance-Plus-Novelty Active Policy

The heuristic active policy selects the next view according to

$$\text{Score}(v) = w_r \cdot \text{Rel}(q, v) + w_n \cdot \text{Nov}(v, S),$$

where

$$\text{Rel}(q, v) = \text{sim}(f_I(v), f_T(q))$$

and

$$\text{Nov}(v, S) = 1 - \max_{u \in S} \text{sim}(f_I(v), f_I(u)).$$

In the main experiments, $w_r = 1.0$ and $w_n = 0.5$. The policy begins from a random initial view and greedily selects the unseen view with the highest score until the budget is reached. Ablations also include a relevance-only variant with $w_n = 0$.

After selection, the observed views are reranked by CLIP similarity and the top-ranked view is returned.

3.7 Learned Reranker

The learned reranker is a two-hidden-layer MLP that takes the concatenation of CLIP text and image features:

$$\text{logit}(q, v) = \text{MLP}([f_T(q); f_I(v)]).$$

The architecture is

$$1024 \to 256 \to \text{ReLU} \to 256 \to \text{ReLU} \to 1.$$

Each (instruction, view) pair is a training example with label 1 if the view is ground truth and 0 otherwise. Training minimises binary cross-entropy with positive class

weighting:

$$\text{pos_weight} = \frac{\#\text{negatives}}{\#\text{positives}}.$$

Only the MLP is trained; CLIP remains frozen. After each epoch, object-level validation accuracy is measured by scoring all five views of each validation sample and checking whether the top-scoring view is ground truth. Early stopping keeps the checkpoint with the best validation accuracy, with patience 10 and a maximum of 50 epochs. Default optimisation uses Adam with learning rate 10^{-3} and batch size 64.

At test time, the learned reranker replaces CLIP cosine similarity when ranking the observed views.

3.8 Trained Active View Selector

The trained active view selector uses the same CLIP+MLP architecture and the same training objective as the learned reranker. The difference is only in how the model is used at test time. During sequential selection, the trained selector scores every unseen view and greedily chooses the highest-scoring next view:

$$v_{t+1} = \arg\max_{v \notin S_t} \text{MLP}([f_T(q); f_I(v)]).$$

After the budget is reached, either CLIP or the learned reranker can be used for final ranking. The main "full pipeline" configuration uses trained selection followed by learned reranking.

This design is deliberately simple. It asks whether the same view-quality signal that is useful for final ranking is also useful for choosing the next view. A limitation, discussed later, is that the selector and reranker therefore learn overlapping objectives.

3.9 Baseline Methods

The following methods are compared:

- **Random**: select remaining views uniformly at random; final ranking uses CLIP.
- **Exhaustive CLIP**: score all candidate views with CLIP and take the global top-1 view. This ignores the budgeted sequential setting and provides an upper reference for pure CLIP ranking.
- **Active policy**: CLIP relevance-plus-novelty selection with CLIP reranking.
- **Learned reranker**: random selection of additional views, then learned MLP reranking of the observed set.
- **Trained active + CLIP**: trained selector with CLIP final ranking.
- **Trained active + learned reranker**: full pipeline.

An earlier fixed-order baseline was removed after dataset shuffling, because view indices no longer carry semantic meaning.

3.10 Evaluation Metric

The main metric is top-1 accuracy under different observation budgets. For each sample, the prediction is correct if the final top-ranked observed view belongs to the ground-truth set. Accuracy over N samples is

$$\text{Accuracy} = \frac{1}{N} \sum_{j=1}^{N} \mathbf{1}(\hat{v}_j \in G_j).$$

Top-2 and Top-3 accuracy are also reported by checking whether any of the top-k ranked observed views is ground truth.

Each stochastic method is evaluated over repeated trials with different random seeds; the final UMD+GT method comparison uses 20 trials, while CO3D runs and ablations use 10 trials. Every trial starts from a random initial view. For ablations, the same initial views are shared across configurations within a trial so that differences are attributable to selection and reranking rather than to different starting points. Means

and standard deviations are reported across trials and shown as error bars on the UMD method-comparison plot.

3.11 Ablation Design

The ablation study measures the contribution of each component relative to a shared baseline:

$$\text{Baseline} = \text{random selection} + \text{CLIP rerank}.$$

Every reported gain is

$$\Delta = \text{accuracy(method)} - \text{accuracy(baseline)}.$$

The compared methods are:

- learned rerank only;
- CLIP-relevance selection;
- CLIP-relevance+novelty selection;
- trained selection;
- CLIP-relevance+novelty selection with learned rerank;
- trained selection with learned rerank (full pipeline).

On CO3D, gains are reported per object category. On UMD+GT, gains are reported per affordance. This makes each plotted value an explicit comparison against the same baseline.

3.12 Implementation Details

The experimental pipeline is implemented in Python with PyTorch and OpenCLIP. The main scripts are:

- `active_view_selection.py`: training, budgeted evaluation, and method comparison plots;
- `build_umd_annotations.py`: conversion of UMD+GT into the common annotation format;
- `evaluate_dataset.py`: schema, split, and ground-truth distribution checks;
- `ablation_study.py`: paired component ablations and gain plots.

All experiments are run on CPU. Features and models are cached so that repeated budget sweeps reuse the same CLIP embeddings and trained weights. Outputs include per-budget JSON summaries, method comparison plots, top-k plots, and ablation heatmaps or bar charts.

3.13 Summary

This chapter presented the methodology and implementation. The project formulates language-guided active view selection as sequential selection under a budget followed by observed-view reranking. CLIP provides frozen image–text features. A heuristic active policy combines relevance and novelty. A learned MLP reranker and a trained selector adapt those features to instruction-specific ground-truth views. The next chapter reports results on the shuffled CO3D dataset and the UMD+GT affordance view-selection dataset.

Chapter 4

Experiments and Results

4.1 Introduction

This chapter presents the experiments and results for the language-guided active view selection system. The aim is to evaluate whether CLIP-based selection, a learned MLP reranker, and a trained active view selector can improve instruction-relevant view prediction under limited observation budgets.

Two datasets are used: the shuffled CO3D multi-view instruction set and the UMD+GT affordance view-selection set. Methods are compared under budgets $k \in \{1, 2, 3, 4, 5\}$. The main UMD+GT comparison uses 20 random-initial trials with mean $\pm$ standard deviation; CO3D results and ablations use 10 trials. Ablation studies measure each component's gain relative to a shared random-plus-CLIP baseline.

4.2 Experimental Setup

Unless otherwise stated, all methods start from a random initial view. Additional views are selected until the budget is reached, after which the observed views are reranked and the top-1 view is returned. A prediction is correct if that view belongs to the

ground-truth set.

The compared methods, matching the legend labels used in the figures, are:

- **Random**: select additional views uniformly at random; rank the observed set with CLIP cosine similarity.
- **Exhaustive CLIP**: score all five candidate views with CLIP and return the global top-1 view (no sequential budget constraint).
- **Active policy**: CLIP relevance-plus-novelty sequential selection, then CLIP ranking of the observed set.
- **Learned reranker**: random additional views, then a trained MLP ranks the observed set.
- **Trained active + CLIP**: a trained MLP selector chooses the next views; CLIP ranks the observed set.
- **Trained active + learned reranker**: full pipeline with trained selection and trained final ranking.

On CO3D, models are trained on objects 16–30, validated on objects 11–15, and evaluated on objects 1–10. On UMD+GT, the official instance split is used, with a held-out validation subset of the training tools. Grouping is by object category on CO3D and by affordance on UMD+GT.

4.3 CO3D Results

4.3.1 Main Method Comparison

Table 4.1 reports mean accuracy on the shuffled CO3D evaluation set. Because views are shuffled, budget 1 no longer benefits from first-view bias: all sequential methods

start near 0.30–0.31, which is close to random chance given the number of ground-truth views per sample.

Table 4.1: Accuracy on the shuffled CO3D evaluation set under different observation budgets.

Method	B1	B2	B3	B4	B5
Random + CLIP	0.300	0.300	0.310	0.300	0.300
Exhaustive CLIP	0.360	0.360	0.360	0.360	0.360
Active policy + CLIP	0.306	0.348	0.356	0.360	0.360
Random + learned reranker	0.306	0.416	0.478	0.498	0.520
Trained active + CLIP	0.306	0.386	0.344	0.368	0.360
Trained active + learned reranker	0.306	**0.534**	**0.520**	**0.520**	**0.520**

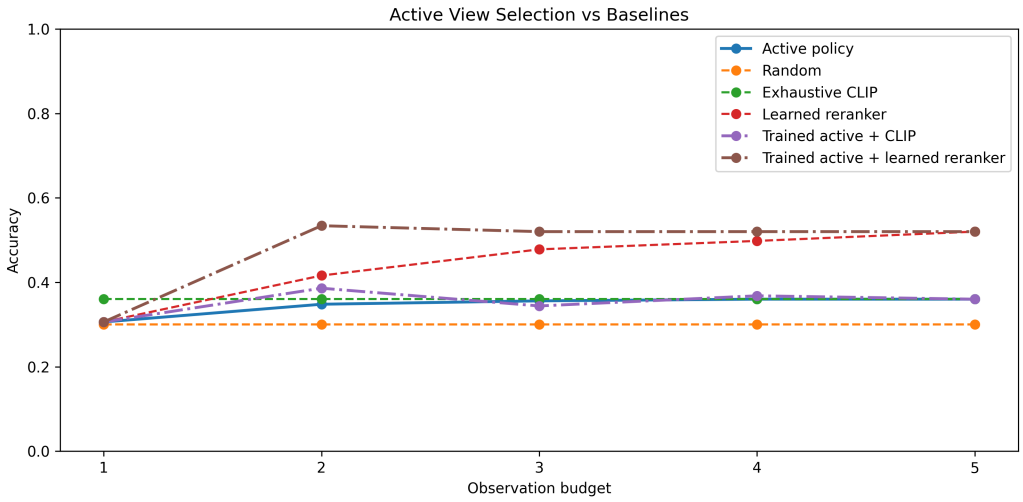


Figure 4.1: CO3D method comparison on the shuffled evaluation set. **Active policy**: CLIP relevance+novelty selection with CLIP final ranking. **Random**: random additional views with CLIP ranking. **Exhaustive CLIP**: global CLIP top-1 over all five views. **Learned reranker**: random selection with MLP final ranking. **Trained active + CLIP**: trained MLP selector with CLIP ranking. **Trained active + learned reranker**: trained selector with MLP ranking (full pipeline).

Figure 4.1 visualises the same comparison. Three findings stand out.

First, CLIP-only methods remain close to exhaustive CLIP. The relevance-plus-novelty active policy reaches about 0.35–0.36 for budgets 2–5, only marginally above the random-plus-CLIP baseline and similar to exhaustive CLIP at 0.36. Language-guided CLIP scoring is therefore better than non-semantic selection, but the heuristic active policy does not clearly beat direct CLIP ranking.

Second, the learned reranker is the main source of improvement. With random selection, learned reranking rises from 0.306 at budget 1 to 0.478 at budget 3 and 0.520 at budget 5. This shows that a lightweight MLP can adapt frozen CLIP features to the instruction-specific ground-truth labels much more effectively than cosine similarity alone.

Third, the full pipeline is strongest at low budgets. At budget 2, trained selection plus learned reranking reaches 0.534, compared with 0.416 for learned reranking alone. By budget 5, both learned-reranker methods converge to 0.520 because the observed set contains all five views and only the final ranking matters.

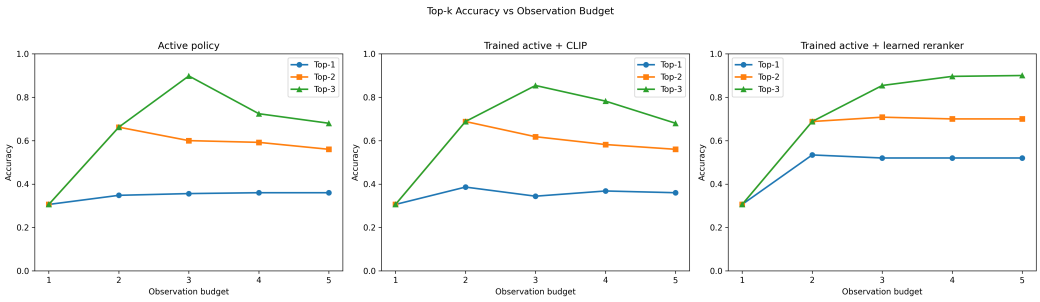


Figure 4.2: CO3D Top-k accuracy versus budget for the active policy. Top-1 is the final predicted view; Top-2/Top-3 succeed if any of the highest-ranked observed views is ground truth.

Figure 4.2 shows Top-1, Top-2, and Top-3 accuracy. Top-2 and Top-3 are consistently higher than Top-1, indicating that correct views are often ranked near the top even when they are not selected as the final answer. This supports the interpretation that ranking quality is the central bottleneck, and that learned reranking addresses it directly.

4.3.2 CO3D Ablation

Figure 4.3 and Table 4.2 report accuracy gains versus the random-plus-CLIP baseline at budget 3. Validation accuracy was 0.80 for the reranker and 0.76 for the selector.

The ablation confirms that learned reranking drives most of the gain. Selection-only methods add only about 3–5 overall points. Interestingly, CLIP relevance-plus-novelty

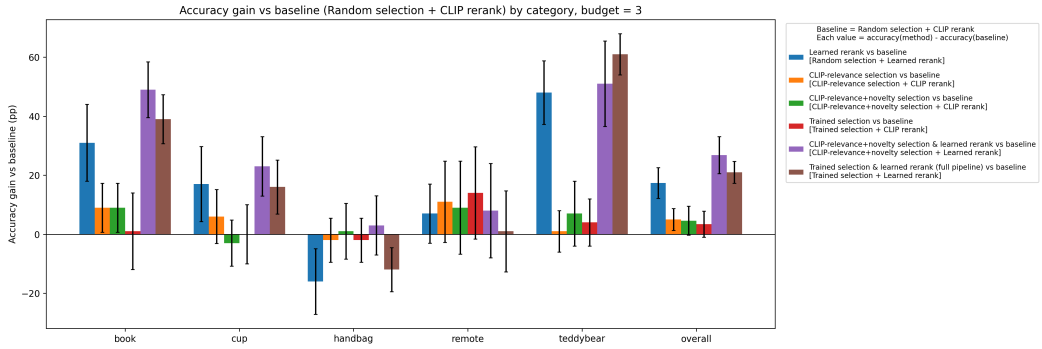


Figure 4.3: CO3D ablation at budget 3. Each bar is accuracy(method) − accuracy(random selection + CLIP rerank). Legend entries name the component being added or replaced relative to that shared baseline (for example, learned rerank only, selection-only variants, and the full trained pipeline).

Table 4.2: CO3D accuracy gain versus random+CLIP at budget 3 (percentage points).

Comparison	book	cup	bag	rem.	teddy	all
Learned rerank	+31.0	+17.0	−16.0	+7.0	+48.0	+17.4
CLIP-relevance selection	+9.0	+6.0	−2.0	+11.0	+1.0	+5.0
CLIP-rel+novelty selection	+9.0	−3.0	+1.0	+9.0	+7.0	+4.6
Trained selection	+1.0	0.0	−2.0	+14.0	+4.0	+3.4
CLIP-rel+nov + learned	+49.0	+23.0	+3.0	+8.0	+51.0	+26.8
Full pipeline	+39.0	+16.0	−12.0	+1.0	+61.0	+21.0

selection combined with learned reranking (+26.8 overall) slightly outperforms the full trained pipeline (+21.0). This suggests that on CO3D the novelty term can supply complementary views to the reranker, whereas the trained selector, which is trained with the same GT-view objective as the reranker, tends to fetch views that the reranker already likes. Category-level results are noisy because each category has only 10 evaluation objects; handbag in particular is negative for several learned-reranker configurations.

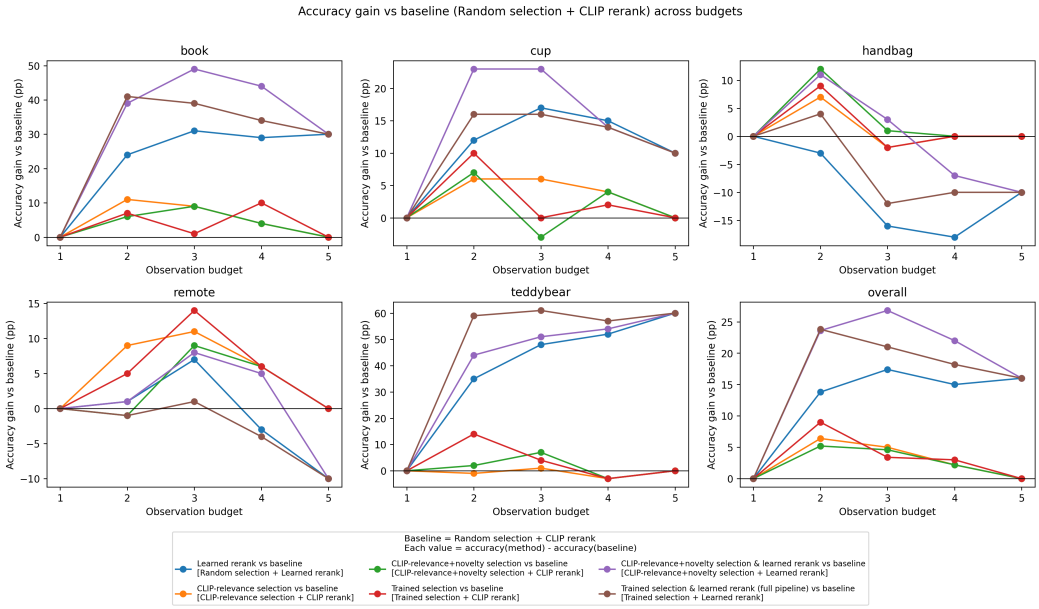


Figure 4.4: CO3D component gains versus budget. Every curve is accuracy(method) − accuracy(random+CLIP), so positive values mean improvement over the shared baseline. Curves correspond to the same component definitions as in Figure 4.3.

4.4 UMD+GT Results

4.4.1 Main Method Comparison

Table 4.3 reports mean accuracy on the UMD+GT test set over 20 random-initial trials. The random-plus-CLIP baseline is about 0.46–0.47, close to the random-pick chance of 0.43 arising from multiple ground-truth views per sample. Exhaustive CLIP reaches only 0.481, showing that cosine similarity alone is a weak final ranker for affordance-

view instructions.

Table 4.3: UMD+GT mean accuracy over 20 trials. Stochastic methods share the same random-initial seeds; values are means (standard deviations appear as error bars in Figure 4.5). At budget 5 the observed set contains all views, so trial variance collapses to zero.

Method	B1	B2	B3	B4	B5
Random + CLIP	0.439	0.465	0.462	0.473	0.481
Exhaustive CLIP	0.481	0.481	0.481	0.481	0.481
Active policy + CLIP	0.439	0.457	0.473	0.484	0.481
Random + learned reranker	0.439	0.571	0.632	0.674	0.696
Trained active + CLIP	0.439	0.524	0.508	0.496	0.481
Trained active + learned reranker	0.439	**0.669**	**0.680**	**0.693**	**0.696**

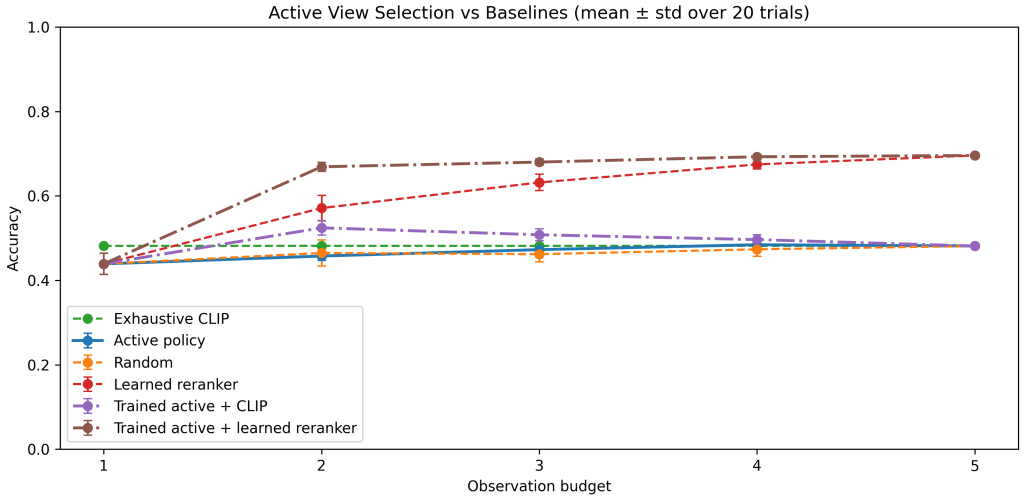


Figure 4.5: UMD+GT method comparison with mean ± std over 20 trials. **Active policy**: CLIP relevance+novelty selection, CLIP final ranking. **Random**: random additional views, CLIP ranking. **Exhaustive CLIP**: global CLIP top-1 over all views (deterministic; no error bar). **Learned reranker**: random selection, MLP final ranking. **Trained active + CLIP**: trained MLP selector, CLIP ranking. **Trained active + learned reranker**: full pipeline (trained selector + MLP ranking).

On UMD+GT, the full pipeline is clearly the strongest method at low and medium budgets. At budget 2 it already reaches 0.669 ± 0.011, while learned reranking alone is 0.571 ± 0.030 and needs about four views to approach the full pipeline. Validation accuracy was 0.744 for the reranker and 0.705 for the selector.

Unlike CO3D, the trained selector provides a clear low-budget benefit when paired with the learned reranker. A likely reason is that UMD turntable sequences change

affordance visibility smoothly with viewpoint, giving the selector a more consistent signal for which unobserved view is likely to be a ground-truth view.

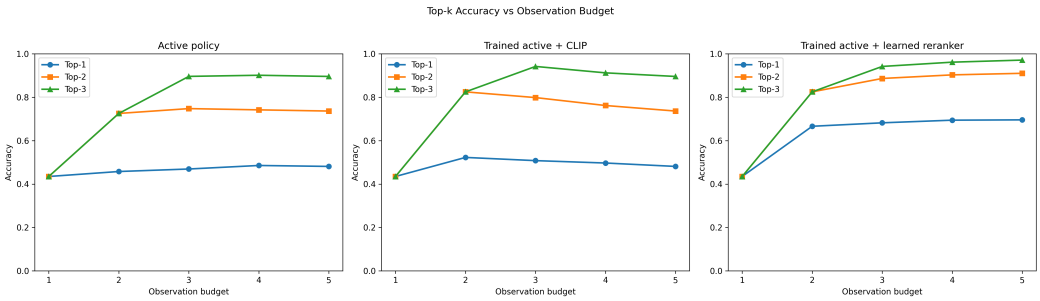


Figure 4.6: UMD+GT Top-k accuracy versus budget. Top-1 is the final predicted view after selection and ranking; Top-2/Top-3 succeed if a ground-truth view appears among the highest-ranked observed views.

4.4.2 UMD+GT Ablation

Figure 4.7 and Table 4.4 report gains versus random-plus-CLIP at budget 3, grouped by affordance.

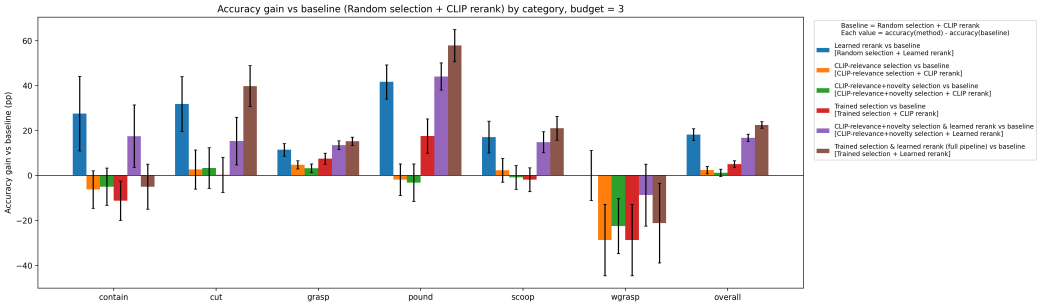


Figure 4.7: UMD+GT ablation at budget 3 by affordance. Each bar is accuracy(method) − accuracy(random selection + CLIP rerank). Legend entries identify the compared component: learned rerank only, CLIP selection variants, trained selection only, combined heuristic selection + learned rerank, and the full pipeline.

The overall ranking at budget 3 is:

1. full pipeline: +22.5 pp;
2. learned rerank only: +18.2 pp;

Table 4.4: UMD+GT accuracy gain versus random+CLIP at budget 3 (percentage points).

Comparison	cont.	cut	grasp	pound	scoop	wgr.	all
Learned rerank	+27.5	+31.8	+11.4	+41.6	+17.1	0.0	+18.2
CLIP-relevance selection	−6.2	+2.7	+4.7	−1.9	+2.3	−28.7	+2.4
CLIP-rel+novelty selection	−5.0	+3.3	+3.2	−3.2	−0.8	−22.5	+1.2
Trained selection	−11.2	+0.2	+7.5	+17.6	−1.9	−28.7	+5.0
CLIP-rel+nov + learned	+17.5	+15.3	+13.5	+44.1	+14.8	−8.8	+16.8
Full pipeline	−5.0	+39.8	+15.2	+57.8	+21.0	−21.2	+22.5

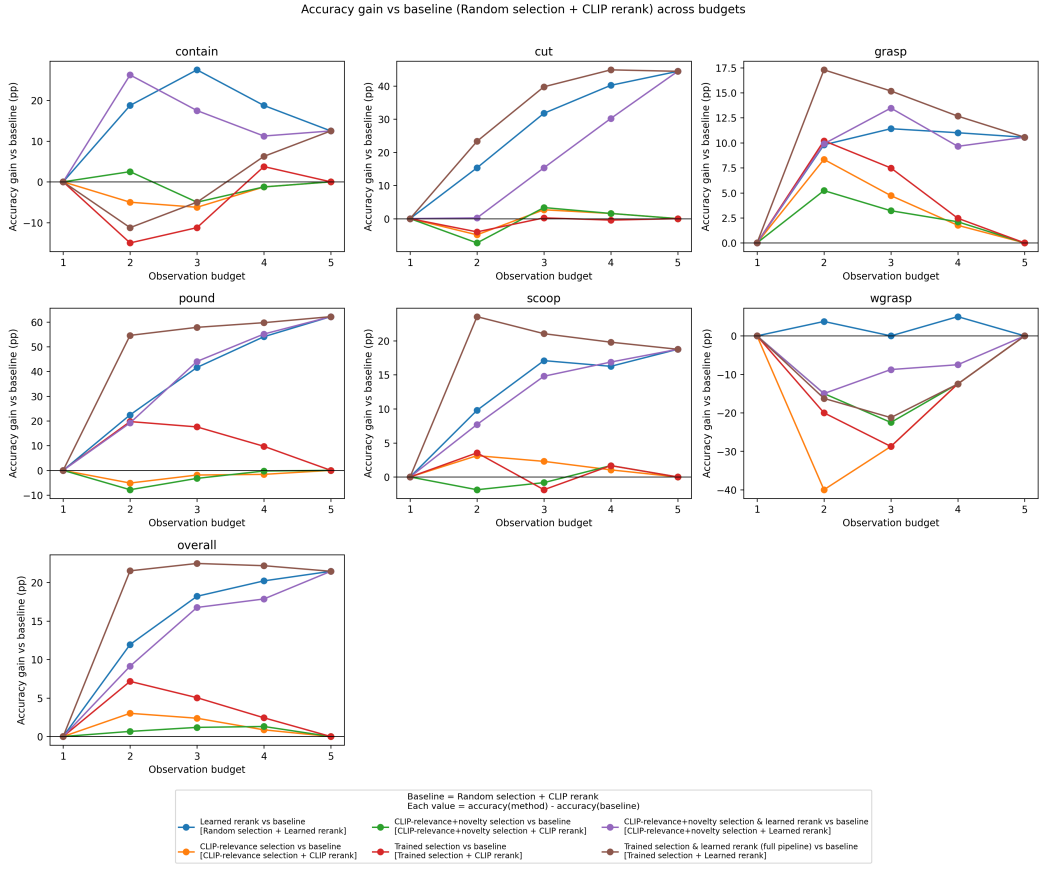


Figure 4.8: UMD+GT component gains versus budget. Every curve is accuracy(method) − accuracy(random+CLIP). Positive values mean the named component configuration beats the shared baseline; the full pipeline (brown) is strongest at low and medium budgets.

3. CLIP-rel+nov + learned rerank: +16.8 pp;
4. trained selection only: +5.0 pp;
5. CLIP selection variants: about +1 to +2 pp.

Pound, cut, and scoop show the largest gains. Wrap-grasp and contain are noisy or negative for several methods; both have only eight test samples after conversion filters remove rotationally ambiguous cases. By budget 5, all learned-reranker methods converge near +16 pp overall, again confirming that selection matters mainly when the observation budget is small.

4.5 Cross-Dataset Discussion

Comparing the two datasets yields a consistent pattern and an important difference.

The consistent pattern is that learned reranking is the dominant improvement over CLIP cosine similarity. On both datasets, CLIP-only active policies stay close to exhaustive CLIP, while replacing the final ranker with the MLP produces large gains as soon as two or more views are available.

The important difference is the value of the trained selector. On shuffled CO3D, CLIP relevance-plus-novelty plus learned reranking can match or exceed the full pipeline. On UMD+GT, the full pipeline is clearly best at budgets 2–4. UMD affordance areas change gradually around the turntable, so the selector can learn a more reliable next-view preference. CO3D instructions such as "front-facing printed design" are more discrete and category-specific, and the selector/reranker objective overlap is more harmful there.

Another practical observation is that budget 1 is identical for all sequential methods: with only the random initial view, selection never occurs and final ranking has only one candidate. Differences therefore appear from budget 2 onward.

4.6 Error Analysis

Several sources of error remain.

First, CLIP cosine similarity is often too coarse for fine-grained viewpoint differences. Exhaustive CLIP stays near 0.36 on CO3D and 0.48 on UMD+GT, only slightly above chance-aware random baselines.

Second, the selector and reranker are trained with the same binary GT-view objective. This makes them redundant: the selector tends to fetch views that the reranker would already score highly. A selector trained to maximise expected improvement given the current observed set could provide complementary behaviour.

Third, both the heuristic and trained selectors score candidate views using their true image features. In a real robot, those images are unavailable until the camera moves. The reported selection results are therefore an optimistic upper bound relative to pose-only next-best-view prediction.

Fourth, ground-truth definitions remain approximate. On CO3D, labels are instruction-dependent and can be subjective. On UMD+GT, relative mask area is a useful proxy for "most clearly visible", but a foreshortened region can cover many pixels without being the most informative view for a human.

Fifth, some groups have few evaluation samples. CO3D categories have 10 evaluation objects each, and UMD contain/wrap-grasp each have only eight test samples. Per-group deltas should therefore be interpreted cautiously.

4.7 Summary

The experiments show that language-guided active view selection is feasible and that a learned MLP on frozen CLIP features substantially improves final view ranking. On CO3D, learned reranking is the main contributor and the full pipeline helps most at low

budgets. On UMD+GT, over 20 trials, the full trained pipeline is the strongest overall method, reaching 0.669 ± 0.011 at budget 2 and 0.680 ± 0.008 at budget 3 against a random-plus-CLIP baseline near 0.46. The next chapter discusses these findings, their limitations, and directions for future work.

Chapter 5

Discussion, Conclusion and Future Work

5.1 Discussion of Results

This project investigated language-guided active view selection for multi-view object perception. The main goal was to test whether CLIP-based scoring, a learned MLP reranker, and a trained active view selector can select views that satisfy a natural language instruction under a limited observation budget.

The results provide clear support for language-guided scoring and stronger support for learned reranking. On both CO3D and UMD+GT, CLIP-only methods outperform non-semantic random selection, but they remain close to exhaustive CLIP. This indicates that image–text similarity is useful, yet cosine similarity alone is not a strong enough final ranker for fine-grained viewpoint instructions.

The learned MLP reranker is the central improvement. By adapting frozen CLIP features to instruction-specific ground-truth views, it raises accuracy substantially as soon as multiple views are observed. On shuffled CO3D, random selection with learned reranking reaches 0.478 at budget 3 and 0.520 at budget 5, compared with about

0.31 for random-plus-CLIP. On UMD+GT, the same pattern is even stronger: learned reranking alone reaches 0.632 at budget 3 and 0.696 at budget 5.

The value of the trained selector depends on the dataset. On UMD+GT, over 20 trials, the full pipeline of trained selection plus learned reranking is clearly best at low budgets, reaching 0.669 ± 0.011 already at budget 2 and 0.680 ± 0.008 at budget 3. Affordance visibility changes smoothly around the turntable, so the selector can learn which next view is likely to be useful. On CO3D, the full pipeline helps at budget 2, but CLIP relevance-plus-novelty combined with learned reranking can match or exceed it at budget 3. This is consistent with the fact that the selector and reranker share the same training objective and therefore tend to reinforce the same view preferences rather than explore complementary ones.

Ablation studies make these contributions explicit. Every plotted gain is defined as accuracy(method) minus accuracy(random selection + CLIP rerank). Across both datasets, selection-only methods add only a few overall points, while any configuration that includes the learned reranker produces double-digit gains. The practical implication is that, for the current problem formulation, improving final observed-view ranking is more important than refining the CLIP heuristic alone.

5.2 Limitations

This project has several limitations.

First, the selector and reranker are trained with the same binary "is this view ground truth?" objective. They therefore learn overlapping signals. A more natural active-perception objective would train the selector to choose the view that most improves the expected final answer given the views already observed.

Second, candidate views are scored using their true image features before they have been "visited". This is acceptable for offline ranking from a pre-collected view set, but it is optimistic for embodied robotics, where the next image is unavailable until the

camera moves. A realistic system would need to predict the value of a candidate pose from the current observation and camera geometry.

Third, both datasets remain modest in scale. CO3D evaluation uses 50 samples (10 per category). UMD+GT has 345 test samples overall, but some affordances such as contain and wrap-grasp have only eight test samples after filters remove rotationally ambiguous cases. Per-group differences are therefore noisy.

Fourth, ground-truth definitions are proxies. CO3D labels depend on human judgement of instruction satisfaction. UMD+GT labels use relative affordance mask area, which correlates with visibility but is not identical to human notions of the most informative viewpoint.

Fifth, the system selects from pre-collected views and does not control a physical robot. It evaluates the decision component of active perception, not continuous camera motion, calibration, or closed-loop control.

Finally, prompt design is fixed. Because the system depends on text embeddings, alternative phrasings of the same instruction could change both CLIP scores and learned-model behaviour.

5.3 Future Work

There are several directions for future work.

First, the selector should be trained with a sequential decision objective. Instead of predicting whether a view is ground truth in isolation, it could estimate the expected accuracy gain of adding a view given the current observed set and the fixed reranker.

Second, next-view prediction should be made pose-conditioned rather than image-oracle. A model that scores candidate camera poses without access to their images would better match real robotic constraints.

Third, the datasets should be expanded. More CO3D categories, more UMD tools after less aggressive filtering, and graded relevance labels would make evaluation more robust.

Fourth, stronger multimodal backbones could be tested. Larger vision-language models with better spatial grounding may improve both zero-shot CLIP baselines and the quality of frozen features used by the MLP.

Fifth, the system could be extended to a real robot. The active policy would then choose the next camera pose, capture a new image, update the observed set, and stop when the budget or a confidence criterion is reached.

Finally, richer language supervision could be studied, including multiple prompt templates, affordance-specific prompts, and instructions that combine viewpoint and action goals such as "find a view suitable for grasping the handle".

5.4 Conclusion

This thesis presented a language-guided active view selection pipeline for multi-view object perception. The system starts from a random view, selects additional views under a limited budget, and reranks the observed views to produce a final prediction. The pipeline includes a CLIP relevance-plus-novelty policy, a learned MLP reranker, and a trained active view selector, evaluated on a shuffled CO3D instruction dataset and an affordance-derived UMD+GT dataset.

The main conclusion is that language-guided scoring is useful, but a lightweight learned reranker on frozen CLIP features is far more effective than CLIP cosine similarity alone. On UMD+GT, combining the trained selector with the learned reranker further improves low-budget performance, reaching about 0.67–0.68 accuracy by budgets 2–3 against a random-plus-CLIP baseline near 0.46, with stable variance across 20 trials. On CO3D, learned reranking remains the dominant contributor, while novelty-aware selection can complement it as well as or better than a selector trained with the same

GT-view objective.

Overall, this project provides a practical experimental foundation for language-guided active perception. With sequential selector training, pose-conditioned next-view prediction, larger datasets, and embodied evaluation, active view selection can become a stronger component of future robotic vision-language systems.

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Appendix: Dataset Structure and Sample Annotations

This appendix provides additional details about the datasets used in the project. The main report describes the methodology and evaluation, while this appendix summarises the common annotation format, the CO3D and UMD+GT organisations, and an example object instance.

A.1 Common Annotation Format

All experiments use a shared JSON annotation format. Each sample contains:

```
{
  "id": "cup::cup_001",
  "category": "cup",
  "instruction": "find the view where the printed design is front-facing",
  "views": [
    "E:/.../view_01.jpg",
    "E:/.../view_02.jpg",
    "E:/.../view_03.jpg",
    "E:/.../view_04.jpg",
    "E:/.../view_05.jpg"
  ],
  "gt_views": [
    "E:/.../view_01.jpg",
    "E:/.../view_05.jpg"
  ]
}
```

UMD samples additionally store `affordance`, `tool`, `split`, and per-view affordance areas. The evaluation code can group results by `category` or by `affordance` and can split by object number or by an explicit `split` field.

A.2 CO3D Folder Structure

The CO3D-derived data is organised by category and object instance:

```
CO3D-subset-<category>/
    <category>/
        <category>_001/
            view_01.jpg
            view_02.jpg
            view_03.jpg
            view_04.jpg
            view_05.jpg
        <category>_002/
            ...
```

Five categories are used: book, cup, handbag, remote, and teddybear. Each has 30 instances and five viewpoints. Combined annotations are stored in `E:/CO3D-combined`. The shuffled annotation file randomises both view order within each sample and sample order in the file, removing first-view bias.

A.3 UMD+GT Conversion Summary

UMD+GT source data contains turntable RGB frames and affordance masks for 104 tools. The conversion script:

- extracts RGB frames only;
- samples five views at 20% increments, with multiple shifted starts per tool;
- creates one sample per affordance present on the tool;
- marks ground-truth views by relative affordance area ($\geq 80\%$ of the maximum);
- drops samples with no clear best view or with negligible affordance area.

The resulting annotation file contains 1,026 samples (603 train / 78 val / 345 test). Random-pick chance accuracy is approximately 0.43.

A.4 Sample CO3D Dataset Entry

Table A.1 shows an example CO3D entry.

Table A.1: Sample CO3D dataset entry for one object instance.

Field	Example
Object category	`cup`
Object instance	`cup_001`
Candidate views	`view_01.jpg`–`view_05.jpg`
Instruction	*find the view where the printed design is front-facing*
Ground-truth views	`view_01.jpg`, `view_05.jpg`
Evaluation budgets	$k = 1, 2, 3, 4, 5$

A.5 Example of Five Candidate Viewpoints

Figure A.1 shows an example object instance with five candidate viewpoints. The active view selection system scores these views against the language instruction and selects a budgeted subset before final reranking.



`view_01` `view_02` `view_03` `view_04` `view_05`

Figure A.1: Example of five candidate viewpoints for one object instance.

A.6 Sample UMD Annotation Entry

Table A.2 shows an example UMD-derived sample.

Table A.2: Sample UMD+GT-derived annotation entry.

Field	Example
Tool / category	`mug_01` / `mug`
Affordance	`grasp`
Instruction	*find the view of the mug where the handle is most clearly visible for grasping*
Split	`test`
GT rule	affordance area $\geq 0.8 \times$ max area